Clarissa Morgan

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Education

California State University, Fresno — Fresno, CA
Bachelor of Science in Computer Engineering — *Aug. 2024 – May 2028*

Lemoore College & Lemoore Middle College High School — Lemoore, CA
Associate's in Science and Mathematics, Associate's in Arts and Humanities, & Diploma — *Aug. 2020 – May 2024*

Experience

Computer Engineering Intern — May 2026 – Aug. 2026
Intel Corporation — *Hillsboro, OR*

- Developed a full-stack system for AI analysis, risk analysis, prediction, and automated insights of Intel data using Intel's internal AI system, Python, SQL, OAuth2, JMT framework, an LLM Gateway, PyUber, SSL verification, Polars dataframes, and Intel's internal DB and DBMS. All data is kept securely inside of Intel's systems throughout the entire process. Usable with any dataset and is entirely customizable. Reduces weekly workload by 75%.
- Developed a full-stack webpage using ScriptHost, PowerBI, TOAD DataPoint, and Docker to analyze and provide actionable insights for tool inventory data. Reduces weekly workload by seven hours.
- Developed a Mass Ping Fab Host Utility with FQDN to IP conversion and optional parallel processing capabilities. Reduces weekly workload by 2 hours.
- Explored and began implementing MCP as the next step for future development of the AI Analytics system.
- Explored RAG capabilities and documented the procedure for utilizing RAG with Intel's internal AI tools. Once complete, this allows for Intel employees to create their own RAG systems for any dataset. Useful for searching through an individual's lengthy and cluttered documentation to quickly find information.

Automotive Ethernet Research Assistant — Jan. 2026 – Present
California State University, Fresno — *Fresno, CA*

- Characterized the voltage profile of intra-vehicular networks of automotive ethernet.
- Engineering cybersecurity measures for automotive ethernet networks.

Artificial Intelligence Research Assistant — May 2025 – July 2025
California State University, Fresno — *Fresno, CA*

- Upgraded detection methods from 2D to 3D vectors to optimize crop harvesting done via a robotic arm (including six DoF). Reduced movement latency by removing iterative physical stepping.
- Identified a critical security vulnerability in the system, which allowed for unauthorized access to the camera feed during power-down states.

Projects

Exploitation & Analysis of Minecraft Server Vulnerabilities | *nginx, OracleVM, hping3* — Jan. 2026 – Present

- Developed a project to exploit and analyze vulnerabilities in Minecraft servers using tools like nginx, OracleVM, hping3, and Minecraft Java Edition.
- Exploited, implemented, and analyzed the affects of packet replay attacks and sustained port attacks (DoS).

Hybrid Counting Sort with Frequency Counting using a Balanced BST | *C++* — Feb. 2026 – May 2026

- Developed an original hybridized sorting algorithm with a time complexity of O(n) with the possibility of a less-than-linear space complexity.

Technical Skills

Languages: C++, ARM, Python, C/C++, SQL (Postgres), DAX, Verilog
Developer Tools: cnvrg.io, Git, Jupyter, Docker, VS Code, TOAD DataPoint, PowerBI, NVIDIA Developer Tools, Intel Quartus Prime Pro, NVIDIA Nsight, Icarus
Hardware and Embedded: FPGA Development - Intel/Altera DE1-SoC, Intel Agilex V - DE25 Standard, Embedded Systems, Embedded OS, UART/I2C, Advanced Circuit Analysis, Oscilloscopes, NVIDIA Jetson AGX Orin, ADCDAC Interfaces.